MAT 2040 Linear Algebra Autumn 2023, CUHK(SZ)

Assignment 6

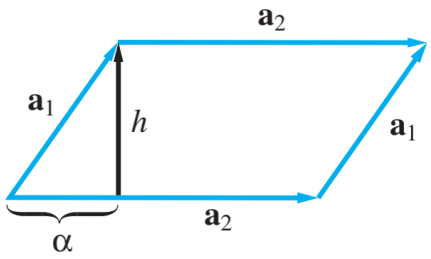
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Please note that

- **Released date**: 11th July
- **Due date**: 21th July, by 11:59 pm.
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX_ ASS7.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)
- Please make sure that your submitted file is clear and readable. Submitted file that can not be opened or not readable will get 0 point.

Question 1. Let $\boldsymbol{A}$ be a 2×2 matrix with linearly independent column vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. If $\mathbf{a}_1$ and $\mathbf{a}_2$ are used to form a parallelogram P with altitude h (see the figure), show that $h^2 \|\mathbf{a}_2\|^2 = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - \left(\mathbf{a}_1^T \mathbf{a}_2\right)^2$



Solution

By the Pythagorean Theorem

$$\alpha^2 + h^2 = \|\mathbf{a}_1\|^2$$

where α is the scalar projection of $\mathbf{a}_1$ into $\mathbf{a}_2$. It follows that

$$\alpha^2 = \frac{(\mathbf{a}_1^T \mathbf{a}_2)^2}{\|\mathbf{a}_2\|^2}$$

and

$$h^2 = \|\mathbf{a}_1\|^2 - \frac{(\mathbf{a}_1^T \mathbf{a}_2)^2}{\|\mathbf{a}_2\|^2}$$

Hence

$$h^2 \|\mathbf{a}_2\|^2 = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - \left(\mathbf{a}_1^T \mathbf{a}_2\right)^2$$

Question 2. Find the constants a and b such that the vectors $\mathbf{u} = \begin{bmatrix} a \\ 3 \\ -b \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ 2 \\ b \end{bmatrix}$ are orthogonal and $a = b + 2$.

Solution

Vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, hence their inner product is equal to zero

$$a^2 + 6 - b^2 = 0.$$

Given that $a = b + 2$, substitute a by $b + 2$ in the above equation

$$(b+2)^2 + 6 - b^2 = 0.$$

Expand the above equation and simplify

$$4b + 10 = 0.$$

Solve for b

$$b = -\frac{5}{2}.$$

Find a

$$a = b + 2 = -\frac{1}{2}$$

Hence, the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $a = -\frac{3}{2}$ and $b = -\frac{1}{2}$.

Question 3. State whether each of the following is true or false:

(a) If the subspaces V and W are orthogonal, then $V^\perp$ and $W^\perp$ are also orthogonal.

(b) If V is orthogonal to W and W orthogonal to Z then V is orthogonal to Z.

Solution

(a) In $\mathbb{R}^n$, suppose that $V = W = \{0\}$. Then since the zero vector is orthogonal to itself, V is orthogonal to $W = V$. However we have $V^\perp = W^\perp = \mathbb{R}^n$ and $\mathbb{R}^n$ is not orthogonal to itself, so $V^\perp$ and $W^\perp$ are not orthogonal. The statement is false.
(b) Suppose that $V = Z = \mathbb{R}^n$ and $W = \{0\}$. Then V is orthogonal to W and W is orthogonal to Z but V is not orthogonal to Z because $\mathbb{R}^n$ is not orthogonal to itself. The statement is false.

Question 4. Let $\mathbf{x}$ and $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^n$ and let $\mathcal{S} = \text{Span}\{\mathbf{x}, \mathbf{y}\}$. Construct a matrix by $\boldsymbol{A} = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$.

(a) Show that $\boldsymbol{A}$ is symmetric.

(b) Show that $\text{Null}(\boldsymbol{A}) = \mathcal{S}^\perp$.

Solution

(a) $\boldsymbol{A}^T = \left(\mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T\right)^T = \left(\mathbf{x}\mathbf{y}^T\right)^T + \left(\mathbf{y}\mathbf{x}^T\right)^T = \mathbf{y}\mathbf{x}^T + \mathbf{x}\mathbf{y}^T = \boldsymbol{A}$.
(b) For any vector $\mathbf{z} \in \mathbb{R}^n$,

$$\boldsymbol{A}\mathbf{z} = \mathbf{x}\mathbf{y}^T\mathbf{z} + \mathbf{y}\mathbf{x}^T\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y},$$

where $c_1 = \mathbf{y}^T\mathbf{z}$ and $c_2 = \mathbf{x}^T\mathbf{z}$. If $\mathbf{z} \in \text{Null}(\boldsymbol{A})$,

$$\mathbf{0} = \boldsymbol{A}\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y}$$

and since $\mathbf{x}$ and $\mathbf{y}$ are linearly independent, we have $c_1 = \mathbf{y}^T\mathbf{z} = \mathbf{0}$ and $c_2 = \mathbf{x}^T\mathbf{z} = 0$. So $\mathbf{z} \perp \mathbf{x}, \mathbf{z} \perp \mathbf{y} \Rightarrow \mathbf{z} \in \mathcal{S}^\perp$. Conversely, if $\mathbf{z} \in \mathcal{S}^\perp$, it follows that

$$\boldsymbol{A}\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y} = \mathbf{0}.$$

Therefore, $\mathbf{z} \in \text{Null}(\boldsymbol{A})$.

Question 5. Let $\mathbf{x} \in \mathbb{R}^n$, $\|\mathbf{x}\|_\infty = \max_{i=1,\cdots,n} |\mathbf{x}_i|$ and $\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |\mathbf{x}_i|^p\right)^{1/p}$.

(a) Show that $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2$.

(b) Show that $\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$.

Solution

(a) According to the definition of $\|\mathbf{x}\|_\infty$ and $\|\mathbf{x}\|_2$, we have

$$\|\mathbf{x}\|_\infty = \max_{i=1,\cdots,n} |\mathbf{x}_i| = \max_{i=1,\cdots,n} \left(|\mathbf{x}_i|^2\right)^{\frac{1}{2}} \leq \left(\sum_{i=1}^{n} |\mathbf{x}_i|^2\right)^{\frac{1}{2}} = \|\mathbf{x}\|_2.$$

(b)

$$\|\mathbf{x}\|_1^2 = \left(\sum_{i=1}^{n} |\mathbf{x}_i|\right)^2 \geq \sum_{i=1}^{n} |\mathbf{x}_i|^2 = \|\mathbf{x}\|_2^2$$

Question 6. An $n \times n$ matrix $\boldsymbol{Q}$ is said to be an orthogonal matrix if the column vectors of $\boldsymbol{Q}$ form an orthonormal set in $\mathbb{R}^n$. Find a, b, c, d such that

$$\boldsymbol{Q} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} & a \\ \frac{1}{2} & -\frac{1}{2} & 0 & b \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} & c \\ \frac{1}{2} & -\frac{1}{2} & 0 & d \end{bmatrix}$$

is an orthogonal matrix.

Solution

If the column vectors form an orthonormal set, then each column in $\boldsymbol{Q}$ is mutually orthogonal and the 2-norm of each column is equal to 1 :

$$\begin{cases} \frac{1}{2}a + \frac{1}{2}b + \frac{1}{2}c + \frac{1}{2}d &= 0 \\ \frac{1}{2}a - \frac{1}{2}b + \frac{1}{2}c - \frac{1}{2}d &= 0 \\ \frac{1}{\sqrt{2}}a - \frac{1}{\sqrt{2}}c &= 0 \\ a^2 + b^2 + c^2 + d^2 &= 1 \end{cases} \Rightarrow \begin{cases} a = 0 \\ b = -\frac{1}{\sqrt{2}} \\ c = 0 \\ d = \frac{1}{\sqrt{2}} \end{cases} \text{ or } \begin{cases} a = 0 \\ b = \frac{1}{\sqrt{2}} \\ c = 0 \\ d = -\frac{1}{\sqrt{2}} \end{cases}$$

Question 7. Find the angle between the vectors $\mathbf{v}$ and $\mathbf{w}$:

$$\mathbf{v} = (2, 1, 3)^T, \ \mathbf{w} = (12, 6, 18)^T$$

Solution

The angle θ between two nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$ satisfies

$$\cos\theta = \frac{\mathbf{v}^T\mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|}, \quad 0 \leq \theta \leq \pi$$
$$= 1$$

So $\theta = 0°$.

Question 8. For the following pair of vectors $\mathbf{x}$ and $\mathbf{y}$, find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$ and verify that $\mathbf{p}$ and $\mathbf{x} - \mathbf{p}$ are orthogonal:

$$\mathbf{x} = (3, 5)^T, \ \mathbf{y} = (1, 1)^T$$

Solution

The vector projection of $\mathbf{x}$ onto $\mathbf{y}$ is

$$\mathbf{p} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y} = (4, 4)^T.$$

Therefore, $\mathbf{x} - \mathbf{p} = (3, 5)^T - (4, 4)^T = (-1, 1)^T$. It is easy to verify that

$$\mathbf{p}^T(\mathbf{x} - \mathbf{p}) = 0 \quad \Rightarrow \ \mathbf{p} \text{ and } \mathbf{x} - \mathbf{p} \text{ are orthogonal.}$$

Question 9. If $\mathbf{x} = (x_1, x_2)^T, \mathbf{y} = (y_1, y_2)^T$, and $\mathbf{z} = (z_1, z_2)^T$ are arbitrary vectors in $\mathbb{R}^2$, prove that:

(a) $\mathbf{x}^T\mathbf{x} \geq 0$

(b) $\mathbf{x}^T\mathbf{y} = \mathbf{y}^T\mathbf{x}$

(c) $\mathbf{x}^T(\mathbf{y} + \mathbf{z}) = \mathbf{x}^T\mathbf{y} + \mathbf{x}^T\mathbf{z}$

Solution

(a) $\mathbf{x}^T\mathbf{x} = x_1^2 + x_2^2 \geq 0$

(b) $\mathbf{x}^T\mathbf{y} = x_1y_1 + x_2y_2 = \mathbf{y}^T\mathbf{x}$

(c) $\mathbf{x}^T(\mathbf{y} + \mathbf{z}) = x_1y_1 + x_1z_1 + x_2y_2 + x_2z_2 = \mathbf{x}^T\mathbf{y} + \mathbf{x}^T\mathbf{z}$

Question 10. (a) Let S be the subspace of $\mathbb{R}^3$ spanned by the vectors $\mathbf{x} = (x_1, x_2, x_3)^T$ and $\mathbf{y} = (y_1, y_2, y_3)^T$. Let

$$\boldsymbol{A} = \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{pmatrix}.$$

Show that $S^\perp = \text{Null}(\boldsymbol{A})$.

(b) Find the orthogonal complement of the subspace of $\mathbb{R}^3$ spanned by $(1, 2, 1)^T$ and $(1, -1, 2)^T$.

Solution

(a) Let $(x, y, z)^T \in S^\perp$. By the definition of orthogonal complement, we have

$$\begin{cases} \mathbf{x}^T(x, y, z)^T = 0 \\ \mathbf{y}^T(x, y, z)^T = 0 \end{cases} \Rightarrow \begin{cases} x_1x + x_2y + x_3z = 0 \\ y_1x + y_2y + y_3z = 0 \end{cases}$$

This system can be written in a matrix form, i.e.,

$$\begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0,$$

which implies $(x, y, z)^T \in \text{Null}(\boldsymbol{A})$. Thus,

$$(x, y, z)^T \in S^\perp \;\Rightarrow\; (x, y, z)^T \in \text{Null}(\boldsymbol{A}).$$

Conversely, we have

$$(x, y, z)^T \in \text{Null}(\boldsymbol{A}) \;\Rightarrow\; (x, y, z)^T \in S^\perp.$$

Therefore, $S^\perp = \text{Null}(\boldsymbol{A})$.

(b) From part (a), we have the orthogonal component of the subspace S is $S^\perp = \text{Null}(\boldsymbol{A})$, where

$$\boldsymbol{A} = \begin{bmatrix} -1 & 2 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

To find the null space of $\boldsymbol{A}$, we first reduce the matrix into echelon form, i.e.,

$$\boldsymbol{A} = \begin{bmatrix} -1 & 2 & 1 \\ 1 & -1 & 0 \end{bmatrix} \xrightarrow{R_2 \to R_2 + R_1} \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Therefore, any $\mathbf{x} = [x_1, x_2, x_3]^T \in \text{Null}(\boldsymbol{A})$ satisfies:

$$\begin{cases} -x_1 + 2x_2 + x_3 = 0 \\ x_2 + x_3 = 0 \end{cases} \Rightarrow \begin{cases} -x_1 + 2x_2 + x_3 = 0 \\ x_3 = -x_2 \end{cases}.$$

Suppose that $x_2 = t$, $t \in \mathbb{R}$, then, $x_3 = -t$. Putting $x_2 = t$, $x_3 = -t$ in the equation $-x_1 + 2x_2 + x_3 = 0$, we have $x_1 = t$.

Therefore, the null space of $\boldsymbol{A}$ is,

$$\begin{aligned} \text{Null}(\boldsymbol{A}) &= \left\{ (x_1, x_2, x_3)^T \right\} \\ &= \left\{ (t, t, -t)^T, t \in \mathbb{R} \right\} \end{aligned}$$

Question 11. For each of the following systems $\boldsymbol{A}\mathbf{x} = \mathbf{b}$, find all least squares solutions:

(a) $\boldsymbol{A} = \begin{pmatrix} 1 & 4 \\ 2 & 1 \\ -1 & -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

(b) $\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 3 \\ -1 & 3 & 2 \\ 1 & 2 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix}$

Solution

Definition: If $\boldsymbol{A}$ is an $m \times n$ matrix of rank n, the normal equations

$$\boldsymbol{A}^T\boldsymbol{A}\mathbf{x} = \boldsymbol{A}^T\mathbf{b}$$

have a unique solution

$$\mathbf{x} = \left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T\mathbf{b}$$

and $\mathbf{x}$ is the unique least squares solution of the system $\boldsymbol{A}\mathbf{x} = \mathbf{b}$.

(a) Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ Substitute the values of $\boldsymbol{A}, \mathbf{x}$, and $\mathbf{b}$ in $\boldsymbol{A}^T\boldsymbol{A}\mathbf{x} = \boldsymbol{A}^T\mathbf{b}$

$$\begin{bmatrix} 1 & 4 \\ 2 & 1 \\ -1 & -3 \end{bmatrix}^T \begin{bmatrix} 1 & 4 \\ 2 & 1 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 2 & 1 \\ -1 & -3 \end{bmatrix}^T \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Solving this linear system, we can obtain

$$x_1 = 19/25,\ x_2 = 4/25$$

(b) Similar with (a), we can solve the linear system to obtain:

$$x_1 = 44\ , x_2 = 32,\ x_3 = -26$$

Question 12. Let $\boldsymbol{P} = \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T$, where $\boldsymbol{A}$ is an $m \times n$ matrix of rank n.

(a) Show that $\boldsymbol{P}^2 = \boldsymbol{P}$.

(b) Prove that $\boldsymbol{P}^k = \boldsymbol{P}$ for $k = 1, 2, \ldots$

(c) Show that $\boldsymbol{P}$ is symmetric. [Hint: If $\boldsymbol{B}$ is non-singular, then $\left(\boldsymbol{B}^{-1}\right)^T = \left(\boldsymbol{B}^T\right)^{-1}$.]

Solution

(a)

$$\begin{aligned}
\boldsymbol{P}^2 &= \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \cdot \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \\
\boldsymbol{P}^2 &= \boldsymbol{A}\left[\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\left(\boldsymbol{A}^T\boldsymbol{A}\right)\right]\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \\
\boldsymbol{P}^2 &= \boldsymbol{A}\left[(\boldsymbol{B})^{-1}(\boldsymbol{B})\right]\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}A^T \quad \left(\text{let } \boldsymbol{B} = \boldsymbol{A}^T\boldsymbol{A}\right) \\
\boldsymbol{P}^2 &= \boldsymbol{A}[\boldsymbol{I}]\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \quad \left(\text{ Since } \boldsymbol{B}\boldsymbol{B}^{-1} = \boldsymbol{I}\right) \\
\boldsymbol{P}^2 &= \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \quad (\text{ Since } \boldsymbol{A}\cdot\boldsymbol{I} = \boldsymbol{A}) \\
\boldsymbol{P}^2 &= \boldsymbol{P}
\end{aligned}$$

(b) The objective is to prove that $\boldsymbol{P}^k = P$, for $k = 1, 2, \ldots$ We prove this by using induction on k. If $k = 1$ then we have $\boldsymbol{P}^1 = P$ Assume $\boldsymbol{P}^m = \boldsymbol{P}$, for some m. Consider

$$\begin{aligned}
\boldsymbol{P}^{m+1} &= \boldsymbol{P}\cdot\boldsymbol{P}^m \\
&= \boldsymbol{P}\cdot\boldsymbol{P} \\
&= \boldsymbol{P}^2 \\
&= \boldsymbol{P}
\end{aligned}$$

Hence the result is true by induction on k.

(c) The objective is to show that $\boldsymbol{P}$ is symmetric. It is enough to prove $\boldsymbol{P}^T = \boldsymbol{P}$. Consider

$$\begin{aligned}
\boldsymbol{P}^T &= \left[\boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T\right]^T \\
&= \left(\boldsymbol{A}^T\right)^T\left[\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\right]^T\boldsymbol{A}^T \\
&= \boldsymbol{A}\left[\left(\boldsymbol{A}^T\boldsymbol{A}\right)^T\right]^{-1}\boldsymbol{A}^T \quad \left(\begin{array}{l}\text{Here } \boldsymbol{A} \text{ is } m\times n \text{ matrix rank } n, \text{ so} \\ \boldsymbol{A}^T\boldsymbol{A} \text{ is non-singular matrix}\end{array}\right) \\
&= \boldsymbol{A}\left[\left(\boldsymbol{A}^T\left(\boldsymbol{A}^T\right)^T\right)\right]^{-1}\boldsymbol{A}^T \\
&= \boldsymbol{A}\left[\left(\boldsymbol{A}^T\boldsymbol{A}\right)\right]^{-1}\boldsymbol{A}^T \\
&= \boldsymbol{P}
\end{aligned}$$

Therefore, $\boldsymbol{P}$ is symmetric.

Question 13. Let $\mathbf{x} = (1,1,1,1)^T$ and $\mathbf{y} = (8,2,2,0)^T$.

(a) Determine the angle θ between $\mathbf{x}$ and $\mathbf{y}$.

(b) Find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$.

(c) Verify that $\mathbf{x} - \mathbf{p}$ is orthogonal to $\mathbf{p}$.

(d) Compute $\|\mathbf{x} - \mathbf{p}\|_2, \|\mathbf{p}\|_2, \|\mathbf{x}\|_2$ and verify that the Pythagorean law is satisfied.

Solution

(a) we know that $\cos\theta = \frac{\langle \mathbf{xy} \rangle}{\|\mathbf{x}\|\|\mathbf{y}\|} = \frac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|} = \frac{12}{\sqrt{4}\sqrt{72}} = \frac{6}{\sqrt{72}} = \frac{6}{2\sqrt{6}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = \pi/4$

(b) The vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$ is

$$\begin{aligned}
\mathbf{p} &= \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y} \\
\mathbf{x}^T\mathbf{y} &= 12,; \ \mathbf{y}^T\mathbf{y} = 72 \\
\mathbf{p} &= \frac{12}{72}\mathbf{y} \\
&= \frac{12}{72}(8,2,2,0)^T \\
&= \left(\frac{4}{3}, \frac{1}{3}, \frac{1}{3}, 0\right)^T
\end{aligned}$$

(c)

$$\begin{aligned}
\mathbf{x} - \mathbf{p} &= (1,1,1,1)^T - \left(\frac{4}{3}, \frac{1}{3}, \frac{1}{3}, 0\right)^T \\
&= \left(\frac{-1}{3}, \frac{2}{3}, \frac{2}{3}, 1\right)^T \\
(\mathbf{x} - \mathbf{p})^T\mathbf{p} &= \frac{-4}{9} + \frac{2}{9} + \frac{2}{9} + 0 \\
&= 0
\end{aligned}$$

(d)

$$\mathbf{x} - \mathbf{p} = \left(\frac{-1}{3}, \frac{2}{3}, \frac{2}{3}, 1\right)^T, \mathbf{p} = \left(\frac{4}{3}, \frac{1}{3}, \frac{1}{3}, 0\right)^T$$

$$\|\mathbf{x} - \mathbf{p}\|_2 = \sqrt{\frac{1}{9} + \frac{4}{9} + \frac{4}{9} + 1} = \sqrt{1+1} = \sqrt{2}$$

$$\|\mathbf{p}\|_2 = \sqrt{\frac{16}{9} + \frac{1}{9} + \frac{1}{9} + 0} = \sqrt{2}$$

$$\|\mathbf{x}\|_2 = \sqrt{1+1+1+1} = \sqrt{4} = 2$$

$$\text{Clearly} \quad \|\mathbf{x} - \mathbf{p}\|^2 + \|P\|^2 = 4 = \|\mathbf{x}\|_2^2$$

Pythagorean law is satisfied.

Question 14. Let

$$\mathbf{u}_1 = \begin{pmatrix} \frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$$

(a) Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis for $\mathbb{R}^3$.

(b) Let $\mathbf{x} = (1,1,1)^T$. Write $\mathbf{x}$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2$, and $\mathbf{u}_3$ using Theorem 21.13 and use Parseval's formula to compute $\|\mathbf{x}\|$.

Solution

(a) We show $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis for $\mathbb{R}^3$, for this we show $\mathbf{u}_1^T\mathbf{u}_1 = 1, \mathbf{u}_1^T\mathbf{u}_2 = 1$

$$\mathbf{u}_3^T\mathbf{u}_3 = 1, \mathbf{u}_1^T\mathbf{u}_2 = 0, \mathbf{u}_1^T\mathbf{u}_3 = 0, \mathbf{u}_2^T\mathbf{u}_3 = 0$$

Consider

$$\begin{aligned}
\mathbf{u}_1^T\mathbf{u}_1 &= \frac{1}{18} + \frac{1}{18} + \frac{16}{18} = 1 \\
\mathbf{u}_2^T\mathbf{u}_2 &= \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = 1 \\
\mathbf{u}_3^T\mathbf{u}_2 &= \frac{1}{2} + \frac{1}{2} + 0 = 1 \\
\mathbf{u}_1^T\mathbf{u}_2 &= \frac{\sqrt{2}}{9} + \frac{\sqrt{2}}{9} - \frac{2\sqrt{2}}{9} = 0 \\
\mathbf{u}_1^T\mathbf{u}_3 &= \frac{1}{6} - \frac{1}{6} + 0 = 0 \\
\mathbf{u}_2^T\mathbf{u}_3 &= \frac{\sqrt{2}}{3} - \frac{\sqrt{2}}{3} + 0 = 0
\end{aligned}$$

The set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ forms an orthogonal basis for $\mathbb{R}^3$.

(b) Let $\mathbf{x} = (1,1,1)^T$. $\mathbf{x}$ as the linear combination of $\mathbf{u}_1, \mathbf{u}_2$ and $\mathbf{u}_3$ is given by $\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3$, where

$$c_1 = \mathbf{x}^T\mathbf{u}_1 = (1,1,1)\begin{bmatrix} \frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} \end{bmatrix} = \frac{1}{3\sqrt{2}} + \frac{1}{3\sqrt{2}} - \frac{4}{3\sqrt{2}} = \frac{-2}{3\sqrt{2}} = \frac{-\sqrt{2}}{3}$$

$$c_2 = \mathbf{x}^T\mathbf{u}_2 = (1,1,1)\begin{bmatrix} 2/3 \\ 2/3 \\ 1/3 \end{bmatrix} = \frac{2}{3} + \frac{2}{3} + \frac{1}{3} = \frac{5}{3}$$

$$c_3 = \mathbf{x}^T\mathbf{u} = (1,1,1)\begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 0 = 0$$

Therefore $\mathbf{x} = -\frac{\sqrt{2}}{3}\mathbf{u}_1 + \frac{5}{3}\mathbf{u}_2, \|\mathbf{x}\| = \sqrt{3}$

Question 15. Let S be the subspace of $\mathbb{R}^3$ spanned by the vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ of Question 14. Let $\mathbf{x} = (1, 2, 2)^T$. Find the projection $\mathbf{p}$ of $\mathbf{x}$ onto S. Show that $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_2$ and $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_3$.

Solution

Projection $\mathbf{p}$ of $\mathbf{x} = (1, 2, 2)^T$ onto S is given by $\mathbf{p} = \boldsymbol{U}\boldsymbol{U}^T\mathbf{x}$ where

$$\boldsymbol{U} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/\sqrt{2} \\ 2/3 & -1/\sqrt{2} \\ 1/3 & 0 \end{bmatrix} \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} =$$

$$\begin{bmatrix} \frac{4}{9} + \frac{1}{2} & \frac{4}{9} - \frac{1}{2} & \frac{2}{9} + 0 \\ \frac{4}{9} - \frac{1}{2} & \frac{4}{9} + \frac{1}{2} & \frac{2}{9} + 0 \\ \frac{2}{9} + 0 & \frac{2}{9} - 0 & \frac{1}{9} + 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{17}{18} & -\frac{1}{18} & \frac{2}{9} \\ -\frac{1}{18} & \frac{17}{18} & \frac{2}{9} \\ \frac{2}{9} & \frac{2}{9} & \frac{1}{9} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 23 \\ 41 \\ 16 \end{bmatrix}$$

As $\mathbf{p} = \dfrac{1}{18} \begin{bmatrix} 23 \\ 41 \\ 16 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, get $\mathbf{p} - \mathbf{x} = \dfrac{1}{18} \begin{bmatrix} 5 \\ 5 \\ -20 \end{bmatrix}$

To show that $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_2$, consider $(\mathbf{p} - \mathbf{x}) \cdot \mathbf{u}_2$

$$\begin{aligned} (\mathbf{p} - \mathbf{x}) \cdot \mathbf{u}_2 &= \frac{1}{18} \begin{bmatrix} 5 \\ 5 \\ -20 \end{bmatrix} \begin{bmatrix} 2/3 \\ 2/3 \\ 1/3 \end{bmatrix} \\ &= \frac{1}{18} \left(\frac{10}{3} + \frac{10}{3} - \frac{20}{3} \right) \\ &= 0 \end{aligned}$$

From the above calculation, we observe that $(\mathbf{p} - \mathbf{x}) \cdot \mathbf{u}_2 = 0$ Therefore, $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_2$

Question 16. Let $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be an orthonormal basis for an inner product space V. If $\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3$ is a vector with the properties $\|\mathbf{x}\| = 5, \langle \mathbf{u}_1, \mathbf{x} \rangle = 4$, and $\mathbf{x} \perp \mathbf{u}_2$, then what are the possible values of c_1, c_2, c_3?

Solution

By Perceval's formula:

$$c_1^2 + c_2^2 + c_3^2 = \|\mathbf{x}\|^2 = 5^2 = 25$$

Consider the following theorem: "Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots \mathbf{u}_n\}$ be an orthonormal basis for an inner product space V. If $\mathbf{v} = \sum_{i=1}^{n} c_i \mathbf{u}_i$ then $c_i = \langle \mathbf{v}_i \mathbf{u}_i \rangle^m$ Use above theorem to find the value of constants as follows:

$$c_1 = \langle \mathbf{u}_1, \mathbf{x} \rangle = 4$$

And

$$c_2 = \langle \mathbf{u}_2, \mathbf{x} \rangle = 0$$

As $c_1^2 + c_2^2 + c_3^2 = 25$, therefore,

$$4^2 + 0^2 + c_3^2 = 25$$
$$c_3^2 = 9$$
$$c_3 = \pm 3$$

Therefore, the values are $c_1 = 4, c_2 = 0, c_3 = \pm 3$

Question 17. Given the basis $\left\{(1, 2, -2)^T, (4, 3, 2)^T, (1, 2, 1)^T\right\}$ for $\mathbb{R}^3$, use the Gram-Schmidt process to obtain an orthonormal basis.

Solution

$\left\{\left(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right)^T, \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)^T, \left(-\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right)^T\right\}$

Question 18. Let $\boldsymbol{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 12 \\ 6 \\ 18 \end{pmatrix}$

(a) Use the Gram-Schmidt process to find an orthonormal basis for the column space of $\boldsymbol{A}$.

(b) Factor $\boldsymbol{A}$ into a product $\boldsymbol{QR}$, where $\boldsymbol{Q}$ has an orthonormal set of column vectors and $\boldsymbol{R}$ is upper triangular.

(c) Solve the least squares problem $A\mathbf{x} = \mathbf{b}$.

Solution

(a) The orthonormal basis is

$$\left\{ \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)^T, \left(\frac{-\sqrt{2}}{6}, \frac{4\sqrt{2}}{6}, \frac{-\sqrt{2}}{6}\right)^T \right\}$$
$$= \left\{ \frac{1}{3}(2,1,2)^T, \frac{\sqrt{2}}{6}(-1,4,-1)^T \right\}$$

(b) We can factorize $\boldsymbol{A}$ into the product $\boldsymbol{\Theta R}$, where

$$\boldsymbol{R} = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix} = \begin{bmatrix} 3 & 5/3 \\ 0 & \sqrt{2}/3 \end{bmatrix}$$
$$\boldsymbol{\Theta} = \begin{bmatrix} 2/3 & -\sqrt{2}/6 \\ 1/3 & 2\sqrt{2}/3 \\ 2/3 & -\sqrt{2}/6 \end{bmatrix}$$

(c) $\boldsymbol{A} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 12 \\ 6 \\ 18 \end{bmatrix}$.

The normal equations for this system are $\boldsymbol{A}^T\boldsymbol{A}\mathbf{x} = \boldsymbol{A}^T\mathbf{b}$

$$\Rightarrow \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 12 \\ 6 \\ 18 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} 9 & 5 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 66 \\ 36 \end{bmatrix}$$
$$9x_1 + 5x_2 = 66$$
$$5x_1 + 3x_2 = 36$$

Multiply first equation with 5 and second equation with 9 and then subtract, we get $x_1 = 9, x_2 = -3$. Therefore $\mathbf{x} = \begin{bmatrix} 9 \\ -3 \end{bmatrix}$.